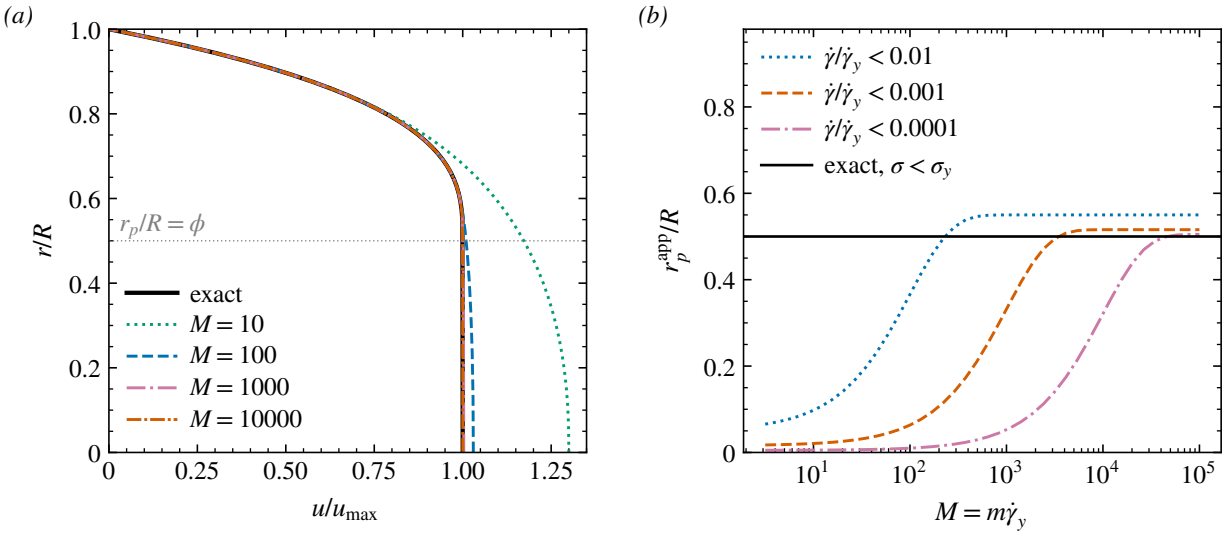
(a)
1.0
0.8
0.6
0.4
0.2
0
r/R
r_p/R = ϕ
exact
M = 10
M = 100
M = 1000
M = 10000
0
0.25
0.50
0.75
1.00
1.25
u/u_max
(b)
γ̇/γ̇_y < 0.01
γ̇/γ̇_y < 0.001
γ̇/γ̇_y < 0.0001
exact, σ < σ_y
r_p^app/R
0.8
0.6
0.4
0.2
0
10^1
10^2
10^3
10^4
10^5
M = mγ̇_y